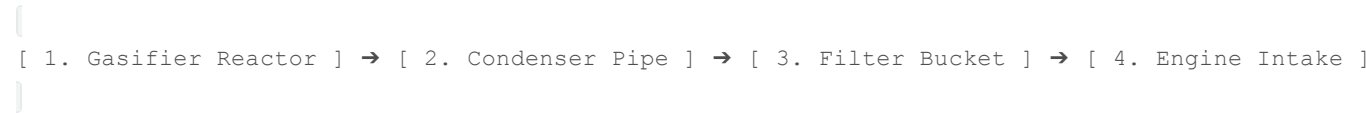


Here is a step-by-step blueprint worksheet to build a classic **"Stratified Downdraft" Gasifier** using your welder, angle grinder, and scrap metal drums. This specific geometry forces the rising resinous smoke from your Sugi wood down through a red-hot charcoal bed, cracking the dangerous tars into clean fuel gas.

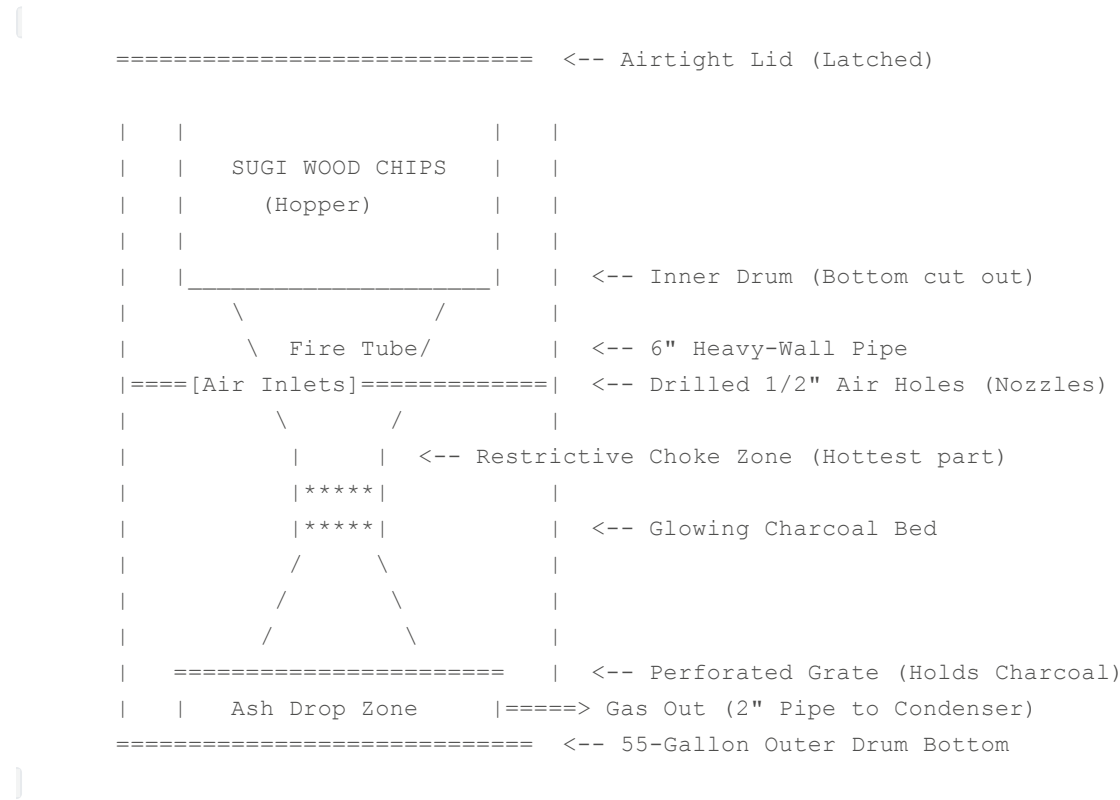
The System Layout (Component Order)

To build this successfully, you must assemble four distinct components in a strict, airtight sequence:



Component 1: The Gasifier Reactor

This is the main chamber where the thermochemical reaction takes place. It is a "tube-within-a-drum" design.



How to build Component 1:

- The Outer Shell:** Take a clean **55-gallon steel oil drum**. Weld a 2-inch pipe coupling near the bottom of the drum. This is your **Gas Out** port.
- The Inner Hopper:** Take a smaller metal drum (like a **20-gallon grease drum**). Cut the bottom completely off so it acts as a funnel.
- The Fire Tube & Choke:** Weld a heavy-walled **6-inch diameter steel pipe** to the bottom opening of the 2-gallon drum. Below this 6-inch pipe, weld a steel reducer cone that tapers down to a **3-inch opening** (this constriction creates the ultra-hot Choke Zone to destroy Sugi tar).

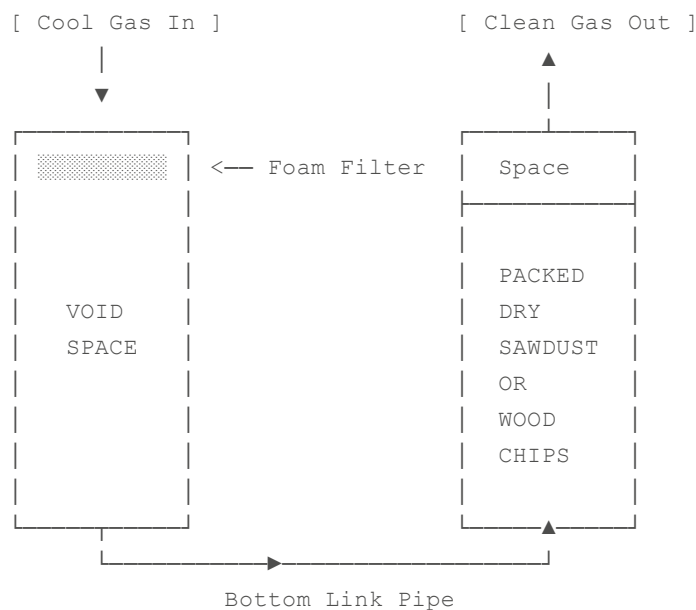
4. **Air Nozzles:** Drill four to six **1/2-inch holes** around the circumference of the 6-inch pipe right above the choke zone. Weld small pipes through the outer drum to feed outside air directly into these holes.
5. **The Grate:** Position a heavy steel grate or perforated plate about 4 inches below the choke zone to hold the charcoal.
6. **Assembly:** Suspend this entire inner hopper assembly inside the 55-gallon drum. Seal the top lid completely with a high-temperature gasket and heavy latches. **It must be 100% airtight.**

* Component 2: The Condenser / Cooler

The gas leaving the reactor will be over 300°C (570°F) and filled with steam from the Sugi wood. Hot gas has low density and won't burn well in your engine.

- **The Build:** Run **10 to 15 feet of 2-inch steel plumbing pipe** in a vertical zigzag pattern (like a radiator) outside the drums.
- **The Drain:** Put a T-fitting with a ball valve at the very lowest point of this pipe layout. This acts as a **condensate drain** to empty out water and moisture before it hits the filter.

🗑 Component 3: The Filter Bucket

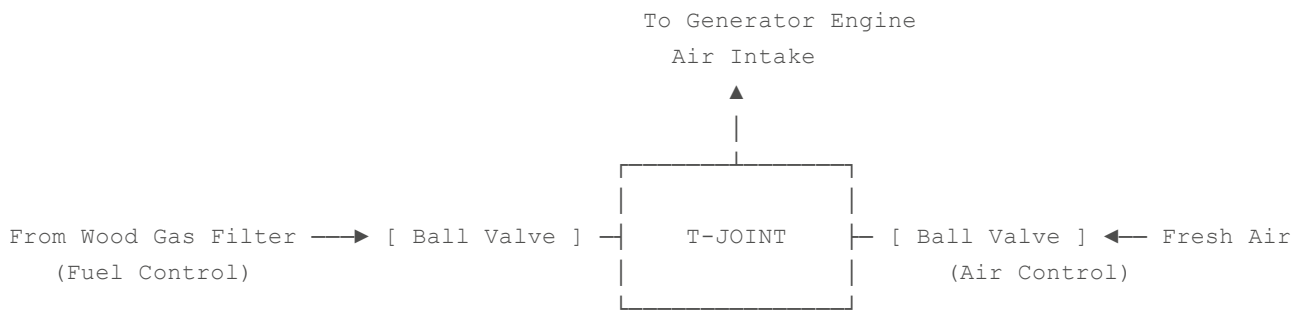


How to build Component 3:

1. Seal a **5-gallon or 10-gallon metal bucket** with an airtight lid.
2. Run the cool pipe from your condenser into the top, dropping down to a void space at the bottom.
3. Pack the upper 3/4 of the bucket tightly with **completely dry Sugi sawdust, wood chips, or a dense foam block**.
4. Pack a final layer of fine cloth or a lawnmower air filter element at the exit hole to catch micro-soot particles.

🔧 Component 4: The Generator Carburetor Interface

Do not modify the inside of your gasoline generator's engine. You only need to build a simple external plumbing T-valve at the air cleaner.



How to build Component 4:

1. Remove the air filter box cover on your gasoline generator.
2. Weld or bolt a **2-inch pipe adapter** directly over the engine's air intake throat.
3. Connect a pipe T-joint to this adapter.
4. On one side of the T-joint, install a ball valve connected to your **Wood Gas Filter output**.
5. On the other side of the T-joint, install a ball valve open to the **Fresh Air**. These two valves act as your manual carburetor.

Worksheet & First-Run Checklist

When you are ready to test your build, follow this sequence exactly:

- **Airtight Check:** Seal all ports, blow air into the system, and spray soapy water on your weld seams. If it bubbles, weld it shut. **Leaks will cause explosions.**
- **The Bed Loading:** Fill the bottom of the reactor grate with raw, real lump charcoal (not briquettes) up to the choke zone.
- **The Fuel Loading:** Dump your bone-dry (<20% moisture) Sugi chips into the top hopper and latch the lid tight.
- **The Flare Test:** Start your generator engine normally on gasoline. Open the wood gas valve slightly to allow the engine's vacuum to pull air through the gasifier. Drop a match or a propane torch into one of the outer air nozzle inlets to ignite the charcoal bed.
- **The Transition:** Watch the flare line. Once you see a thick, clean, yellowish-clear gas instead of white soot, slowly close the generator's gasoline fuel valve while simultaneously opening the wood gas valve and adjusting the fresh air valve until the engine stabilizes.

That **45-liter old gas cylinder (gas bomb)** is absolutely perfect for the internal **Fire Tube / Reaction Hearth**. Because it is made of thick-walled, heavy-duty pressure-rated steel, it will easily withstand the intense 900°C–1000°C (1650°F–1830°F) temperatures required to crack the sticky tars in your Sugi wood, whereas thin sheet metal would warp or burn through in a few hours.

Using a 45L cylinder nested inside a standard 55-gallon (approx. 200-liter) drum gives you the ideal structural proportions for a **10kW capacity system**.

⚠ CRITICAL SAFETY WARNING BEFORE CUTTING THE CYLINDER

Do not touch that gas cylinder with an angle grinder or welding torch until it is 100% purged. Old gas bombs retain explosive vapor inside the pores of the steel even if they have been empty for years.

1. Remove the main valve completely.
2. Fill the cylinder to the absolute brim with **water and liquid dish soap** to displace all residual gas.
3. Keep it completely full of water while making your very first puncture cut, or flush it thoroughly with exhaust fumes from a running vehicle to displace all oxygen.

🔧 Modifying Your 45L Gas Cylinder

You will use your angle grinder to transform the thick cylinder into the primary inner core of the gasifier.



Step-by-Step Cylinder Preparation:

1. **Chop Off the Top:** Cut the top dome/cup off the cylinder where it starts to round out. This wide-open top will face upward to receive the Sugi wood chips from the outer drum lid.
2. **Utilize the Built-In Taper:** The natural neck-down profile at the bottom of the gas cylinder is excellent. It acts as a built-in funnel to direct the wood chips toward the hot zone.
3. **Drill the Air Nozzles:** Measure roughly 2 inches (50mm) above where the bottom constriction begins. Drill **six 1/2-inch (13mm) holes** evenly spaced around the circumference of the cylinder.
4. **Weld the Air Intake Pipes:** Weld a short 3-inch long piece of steel pipe over each of these holes. These will line up with matching holes you drill in the outer 55-gallon drum, allowing fresh air to be sucked directly into the heart of the fire.
5. **Fabricate the Choke Zone:** Inside the neck of the cylinder, weld a heavy-duty steel pipe reducer cone that tapers the opening down to exactly **3 inches (76mm) in diameter**. This narrow gap restricts the flow, creating a super-heated restriction zone that forces the volatile Sugi resins to burn cleanly into gas.

6. **Open the Bottom:** Cut off the very bottom base ring of the cylinder so that ash and charcoal can drop freely onto your lower collection grate.
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Assembly: Nesting the Cylinder inside the 55-Gallon Drum

Once your gas cylinder core is prepped, you will hang it vertically inside the thin-walled 55-gallon drum.

- **Suspension:** Weld steel support brackets near the top rim of the 45L cylinder and attach them to the top inner rim of the 55-gallon drum so the cylinder hangs perfectly centered.
 - **The Nozzle Pass-Throughs:** Line up the air nozzles on your inner cylinder with holes drilled through the outer drum. Use flexible high-temperature appliance conduit or steel pipe to seal the gap between the inner core and the outer drum shell. **The air entering the nozzles must come from the outside, not from inside the outer drum.**
 - **The Bottom Space:** Ensure there is at least 6 to 8 inches (150mm–200mm) of empty space between the bottom of your modified gas cylinder and the bottom floor of the 55-gallon drum. This gives you space to build a heavy steel grate to support your starter charcoal bed and collect ash.
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How would you like to handle the **airtight top seal** for the 55-gallon drum? I can give you instructions on how to use a **bolt-ring clamp lid with a high-temperature silicone gasket**, or show you how to weld a permanent flange system.